

Electrical Circuit Analysis

Time: 3 hours

Answer All Questions

Max. Marks: 60

PART-A

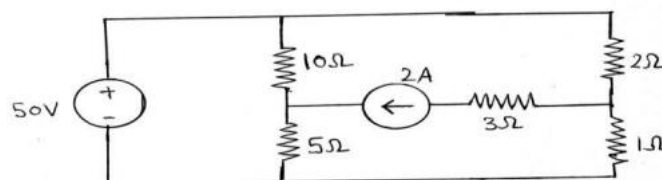
10 X 1M = 10M

	CO	BTL
1. a. State Ohm's law and specify the limitations of ohm's law.	1	1
b. List out the types of network reduction techniques.	1	1
c. Recall phase sequence.	1	1
d. State reciprocity theorem.	1	1
e. Define the time constant of RL circuit.	5	1
f. Why is parallel resonance circuit called rejecter circuit?	2	2
g. Define Tie-set and mention how it is called as?	3	1
h. Write down the condition for reciprocity in terms of ABCD and hybrid parameters?	4	1
i. Draw the transient growth and decay curves for an L –R circuit.	5	1
j. Given the lower and higher cut-off frequency of a band-pass filter are 2.5kHz and 10kHz. Determine its bandwidth.	6	3

PART-B

5 X 10M = 50M

			CO	BTL	
2.	a.	What are the types of sources? Explain them with suitable diagrams and Characteristics?	5M	1	2
	b.	Determine the current in the 5Ω resistor in the network given in figure	5M	1	4

**OR**

3. a. Derive the relation between phase and line values of a 3-phase balanced delta connected system.	5M	1	3
b. Three impedances each of $(10+j3)$ ohms are connected in star to a 220 V, 3-phase, 50 Hz supply. Calculate the line currents and power delivered to the load.	5M	1	4
4. a. Explain the measurement of power in a balanced 3-phase system using a single watt meter	5M	1	2
b. Three impedances each of $(10+j3)$ ohms are connected in star to a 220 V, 3-phase, 50 Hz supply. Calculate the line currents and power delivered to the load	5M	1	4

OR

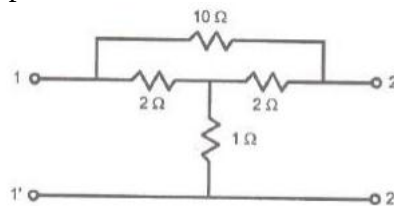
5. Derive the condition for the maximum power to be transferred from the source to the load with example.	10M	1	3
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|----|----|---|----|---|---|
| 6. | a. | Explain the procedure to draw the locus diagram of R-L series circuit when L is varying. | 5M | 2 | 2 |
| | b. | Determine the resonant frequency and quality factor of a coil for the series circuit consisting of $R=10\ \Omega$, $L = 0.1\text{H}$ and $C=10\ \mu\text{F}$ | 5M | 2 | 4 |

OR

- | | | | | | |
|----|--|--|-----|---|---|
| 7. | | A series RLC circuit has to be designed so that it has a bandwidth of 320 Hz and inductance of the coil is 0.2H. It is has to resonate at 350Hz, determine the resistance of coil and capacitance of condenser. If the applied voltage is 150V, determine the voltage across capacitor and coil. | 10M | 2 | 4 |
|----|--|--|-----|---|---|

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|----|----|---|----|---|---|
| 8. | a. | Obtain the ABCD parameters of the circuit shown below | 5M | 4 | 3 |
|----|----|---|----|---|---|



- | | | | | | |
|--|----|--|----|---|---|
| | b. | Derive the relationship between hybrid and Z parameters of two port network. | 5M | 4 | 3 |
|--|----|--|----|---|---|

OR

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|----|----|---|----|---|---|
| 9. | a. | Discuss the various types of network topology matrices which are useful for solving any electric circuit or network problem by using their equivalent graphs. | 7M | 3 | 4 |
|----|----|---|----|---|---|

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|--|----|-----------------------------------|----|---|---|
| | b. | Outline the principle of duality. | 3M | 3 | 4 |
|--|----|-----------------------------------|----|---|---|

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|-----|----|---|----|---|---|
| 10. | a. | Explain about the transient response of series RL circuit to the AC excitation for zero initial conditions. | 5M | 5 | 2 |
|-----|----|---|----|---|---|

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|--|----|--|----|---|---|
| | b. | Derive the expression for the current in a series RL circuit ($R = 10\Omega$, $L = 10\text{mH}$) excited by a sinusoidal voltage of 100V, 50 Hz if the supply is connected at $t = 0$. Assume zero initial conditions. | 5M | 5 | 3 |
|--|----|--|----|---|---|

OR

- | | | | | | |
|-----|--|---|-----|---|---|
| 11. | | Derive the expression to determine the characteristic impedance of constant K High Pass Filter. | 10M | 6 | 3 |
|-----|--|---|-----|---|---|

Electrical Circuit Analysis

SCHEME OF EVALUATION

PART-A

- 1 a. State Ohm's law and specify the limitations of ohm's law. – **1M**
b. List out the types of network reduction techniques. – **1M**
c. Recall phase sequence. – **1M**
d. State reciprocity theorem. – **1M**
e. Define the time constant of RL circuit. – **1M**
f. Why is parallel resonance circuit called rejecter circuit? – **1M**
g. Define Tie-set and mention how it is called as? – **1M**
h. Write down the condition for reciprocity in terms of ABCD and hybrid parameters? – **1M**
i. Draw the transient growth and decay curves for an L –R circuit. – **1M**
j. Given the lower and higher cut-off frequency of a band-pass filter are 2.5kHz and 10kHz. Determine its bandwidth. – **1M**

PART-B

- 2 a Types -2M
Explanation – 2M
Diagrams- 1M
b Loop equations or Node equations – 2M
Solving- 2M
Ans- 1M
- 3 a Relationship between voltages – 2M
Relationship between currents – 2M
Diagrams- 1M
b Formulas – 2M
Solving- 2M
Ans- 1M
- 4 a Explanation – 3M
Diagrams- 2M
b Formulas – 2M
Solving- 2M
Ans- 1M

- 5 Statement and derivation – 6M
 Example- 4M
- 6 a Explanation – 3M
 Diagrams- 2M
 b Formulas – 2M
 Solving- 2M
 Ans- 1M
- 7 Formulas – 5M
 Solving- 4M
 Ans- 1M
- 8 a Solution – 5 M (Each step 1 Mark)
 b Solution – 5 M (Each Conversion 2.5 Marks)
- 9 a Incidence matrix- 2M
 Tie set Matrix – 2M
 Cut set Matrix – 3M
 b Explanation- 3M
- 10 a Derivation – 5M (Each step 1 Mark)
 b Formulas – 2M
 Solving- 2M
 Ans- 1M
- 11 Derivation – 10M
 (T- Type 5 Marks)
 (II- Type 5 Marks)

Electrical Circuit Analysis (20EE21003) KEY
PART-A.

1a. State Ohm's law and specify the limitations of ohm's law. – 1M

Ans:

Ohm's Law

Ohm's Law states that the current (I) flowing through a conductor between two points is directly proportional to the voltage (V) across the two points, provided the physical conditions such as temperature remain constant. Mathematically, it is expressed as: $V=I \cdot R$

Limitations of Ohm's Law

Ohm's Law is not universally applicable and has the following limitations:

1. Temperature Dependence
2. Non-Linear Materials
3. Non-Ohmic Devices
4. Time-Varying Systems

1b. List out the types of network reduction techniques. – 1M

Ans:

1. Series and Parallel Combination of Resistors
2. Delta-to-Star (Δ -Y) and Star-to-Delta (Y- Δ) Transformation

1c. Recall phase sequence. – 1M

Ans:

Phase sequence refers to the order in which the voltages or currents in a three-phase system reach their maximum positive value (peak) at their respective terminals. It is an essential concept in the operation of three-phase AC systems, particularly in power systems and machinery.

Types of Phase Sequence

1. **Positive Sequence (ABC or RYB):**
 - The three-phase voltages reach their maximum in the order A-B-C (or R-Y-B).
 - This is the standard phase sequence used in most systems.
2. **Negative Sequence (ACB or RBY):**
 - The three-phase voltages reach their maximum in the order A-C-B (or R-B-Y).
 - This sequence is the reverse of the positive phase sequence and is rarely used intentionally.

1d. State reciprocity theorem. -1M

Ans:

In a linear, bilateral network with a single independent voltage or current source, the response (current or voltage) at any branch due to a source in another branch is the same as the response at the original source branch if the source and response are interchanged.

1e. Define the time constant of RL circuit. - 1M

Ans:

The time constant of an RL circuit (τ) is the time it takes for the current in the circuit to change significantly after a change in voltage is applied. It is defined as the time required for the current to reach approximately 63.2% of its final steady-state value in a series RL circuit during a transient process, or to decay to 36.8% of its initial value during discharging.

The time constant (τ) of an RL circuit is given by:

$$\tau = \frac{L}{R}$$

1f. Why is parallel resonance circuit called rejecter circuit? -1M

Ans:

A parallel resonance circuit is called a rejecter circuit because at resonance, it exhibits a very high impedance to the resonant frequency and thus "rejects" the flow of current at that specific frequency.

1g. Define Tie-set and mention how it is called as? -1M

Ans:

A tie-set (also known as a fundamental loop) is a closed path in a connected electrical network that includes one branch from the spanning tree of the network. The tie-set is formed by combining one tree branch with a set of links (branches not in the tree) in such a way that the result forms a loop.

1h. Write down the condition for reciprocity in terms of ABCD and hybrid parameters? -1M

Ans:

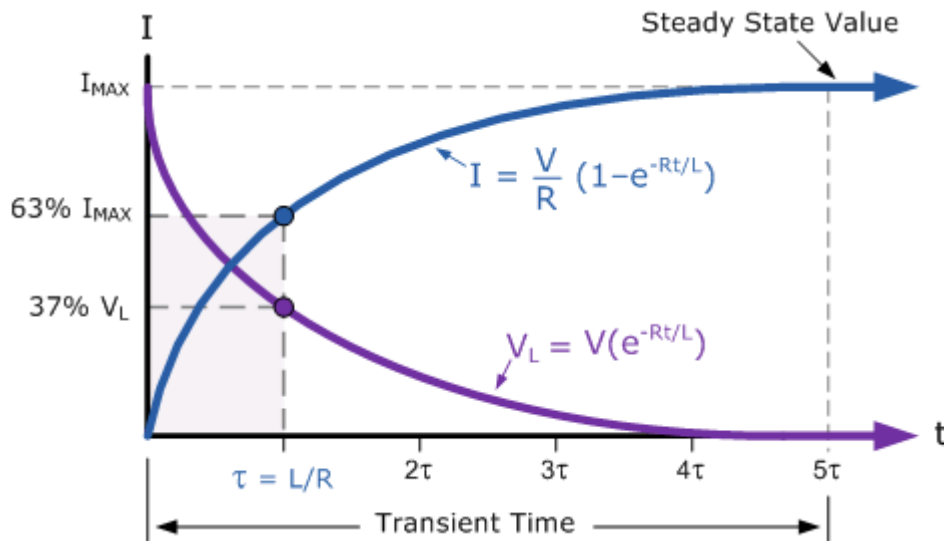
Condition for reciprocity:

ABCD Parameters: **$AD - BC = 1$**

h- Parameters: **$h_{12} = -h_{21}$**

1i. Draw the transient growth and decay curves for an L –R circuit. – 1M

Ans:



1j. Given the lower and higher cut-off frequency of a band-pass filter are 2.5kHz and 10kHz. Determine its bandwidth. - 1M

Ans:

Given:

$$f_L = 2.5 \text{ kHz}, f_H = 10 \text{ kHz}$$

The bandwidth BW is calculated as:

$$BW = f_H - f_L$$

Substituting the given values:

$$BW = 10 \text{ kHz} - 2.5 \text{ kHz} = 7.5 \text{ kHz}$$

So, the bandwidth of the band-pass filter is 7.5 kHz.

2a. What are the types of sources? Explain them with suitable diagrams and Characteristics? -5M

Ans:

CLASSIFICATION OF SOURCES

▪ The sources are classified into 2 types :

1. **INDEPENDENT SOURCE** : An independent source is a source that is not affected by any other circuit element. Examples of independent sources include batteries and generators.
2. **DEPENDENT SOURCE**: A dependent source is a source that is controlled by another circuit element. Examples of dependent sources include voltage-controlled voltage sources and current-controlled current sources.

INDEPENDENT SOURCE:

Independent source is classified into 2 types:

1. **VOLTAGE SOURCE** :A voltage source whose output voltage does not depend upon the voltage or current of any other part of the circuit is known as an independent voltage source.

Classification:

1. **IDEAL VOLTAGE SOURCE** : For ideal voltage source the internal resistance should be 0(zero) and the condition is load voltage is equal to source voltage.
2. **PRACTICAL VOLTAGE SOURCE** :For the practical Volt source having small internal resistance and condition is load Voltage is less than the source voltage.

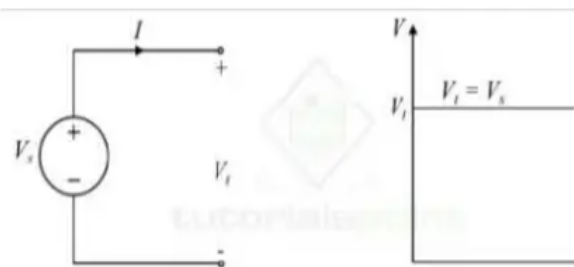


Figure - Ideal Voltage Source

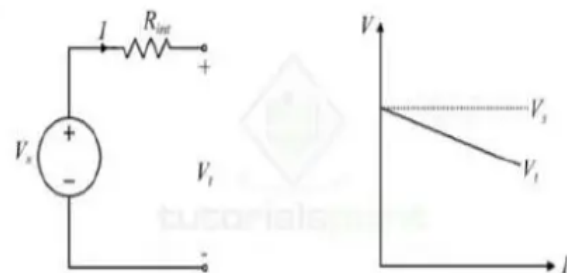


Figure - Practical Voltage Source

2. **CURRENT SOURCE**: The type of current source whose output current does not depend upon the voltage or current of any other part of the circuit is known as **independent current source**.

Classification :

1. **IDEAL CURRENT SOURCE**: For the ideal current source the internal resistance should be infinity and the condition is load current is equal to source current.
2. **PRACTICAL CURRENT SOURCE**: For practical current source having high internal resistance and the condition is load current is less than the source current.

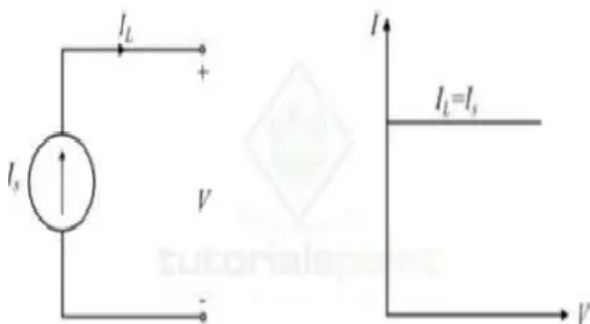


Figure - Ideal Current Source

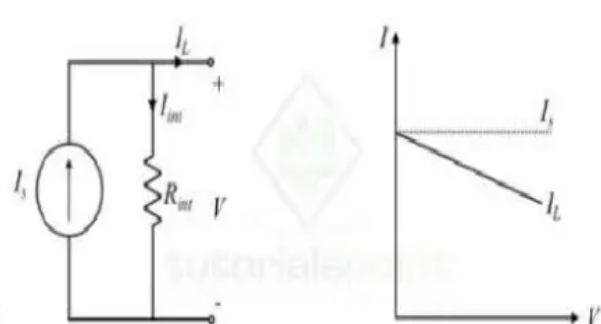


Figure - Practical Current Source

DEPENDENT SOURCE :

- Dependent sources are those whose output voltage is governed by a voltage or current in any other part of the circuit. Dependent sources are represented using diamond shapes as shown in the figure below

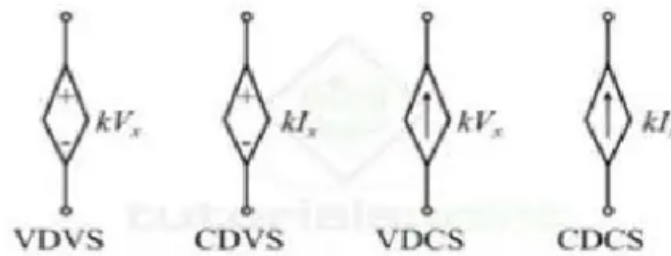
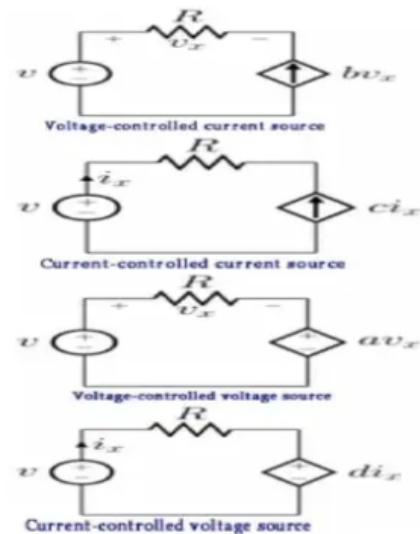


Figure - Dependent (Controlled) Sources

dependent source is classified into 4 types:

- VOLTAGE DEPENDENT VOLTAGE SOURCE:**
The voltage depends on some another circuit of the voltage source.
- VOLTAGE DEPENDENT CURRENT SOURCE:**
The current source is dependent on the some other circuit of the voltage source.
- CURRENT DEPENDENT CURRENT SOURCE:**
The current source is dependent on the some another circuit of current source.
- CURRENT DEPENDENT VOLTAGE SOURCE:**
The voltage source is depends on the some another circuit of the current source.



2b. Determine the current in the 5Ω resistor in the network given in figure. - 5M

Ans:

2b

loop analysis

Loop 1: $50 - 10(i_1 - i_2) - 5(i_1 - i_3) = 0$
 $15i_1 - 10i_2 - 5i_3 = 50 \quad \text{--- (1)}$

Super mesh & Loop 2 & Loop 3
 $-10(i_2 - i_1) - 2i_2 - i_3 - 5(i_3 - i_1) = 0$
 $15i_1 - 12i_3 - 6i_3 = 0 \quad \text{--- (2)}$

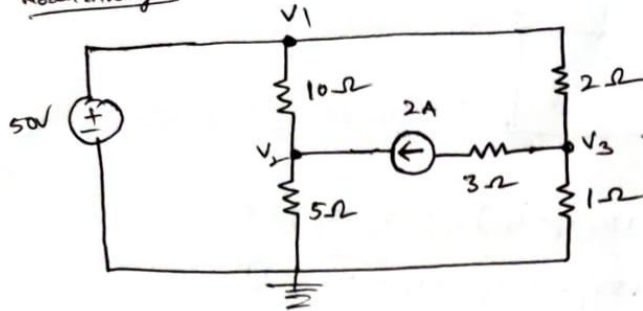
from c.s $i_2 - i_3 = 2 \quad \text{--- (3)}$

$i_1 = 20A, \quad i_2 = 17.33A, \quad i_3 = 15.33A$

$\therefore i_{5\Omega} = i_1 - i_3$
 $= \underline{4.67A}$

2(b)

Node Analysis



Node 1:

$$V_1 = 50V$$

Node 2:

$$\frac{V_2 - V_1}{10} + \frac{V_2 - 0}{5} - 2 = 0$$

$$5(V_2 - V_1) + 10V_2 - 100 = 0$$

$$-5V_1 + 15V_2 = 100 \quad \text{--- (2)}$$

Node 3:

$$\frac{V_3 - V_1}{2} + \frac{V_3}{1} + 2 = 0$$

$$V_3 - V_1 + 2V_3 + 4 = 0$$

$$V_1 - 3V_3 = 4 \quad \text{--- (3)}$$

From eq (2) $-5V_1 + 15V_2 = 100$

$$-5(50) + 15V_2 = 100$$

$$15V_2 = 100 + 250 = 350$$

$$V_2 = 23.33V$$

$$I_{5\Omega} = \frac{V_2}{5}$$

$$= \underline{\underline{4.67A}}$$

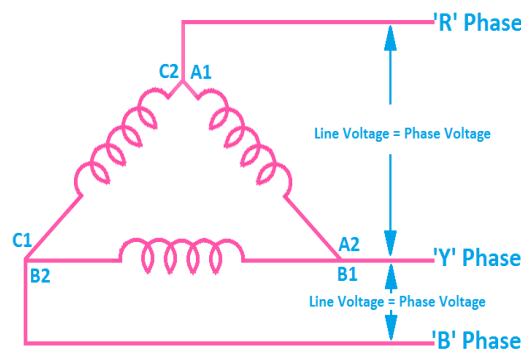
3a. Derive the relation between phase and line values of a 3-phase balanced delta connected system. **-5M**

Ans:

Delta or Mesh Connection (Δ) System is also known as **Three Phase Three Wire System (3-Phase 3 Wire)** and it is the most preferred system for AC power transmission while for distribution, Star connection is generally used.

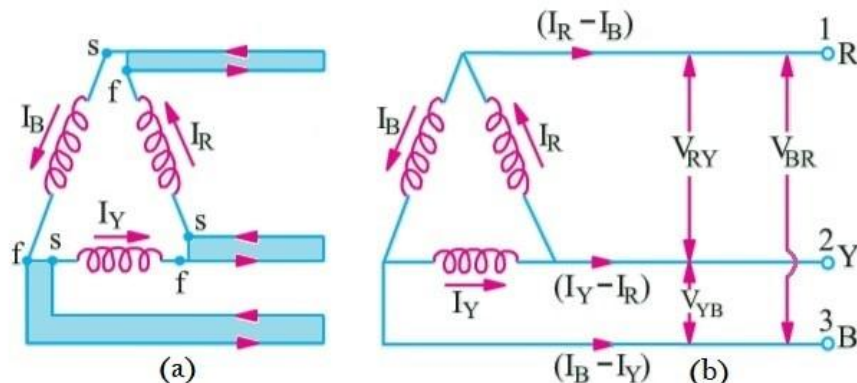
In **Delta** (also denoted by Δ) system of interconnection, the starting ends of the three phases or coils are connected to the finishing ends of the coil. Or the starting end of the first coil is connected to the finishing end of the second coil and so on (for all three coils) and it looks like a closed mesh.

In more clear words, all three coils are connected in series to form a close mesh or circuit. Three wires are taken out from three junctions and the all outgoing currents from junction assumed to be positive.



In **Delta connection**, the three windings interconnection looks like a short circuit, but this is not true, **if the system is balanced, then the value of the algebraic sum of all voltages around the mesh is zero** in Delta connection. When a terminal is open in Δ , then there is no chance of flowing currents with basic frequency around the closed mesh.

In Delta configuration, at any instant, the EMF value of one phase is equal to the resultant of the other two phases EMF values but in the opposite direction.



Voltage and Current in Delta Connection (Δ)

Line Voltages (V_L) and Phase Voltages (V_{Ph}) in Delta Connection

It is seen in fig that there is only one phase winding between two terminals (i.e. there is one phase winding between two wires). Therefore, **in Delta Connection, the voltage between (any pair of) two lines is equal to the phase voltage of the phase winding** which is connected between two lines.

Since the phase sequence is $R \rightarrow Y \rightarrow B$, therefore, the direction of voltage from R phase towards Y phase is positive (+), and the voltage of R phase is leading by 120° from Y phase voltage. Likewise, the voltage of Y phase is leading by 120° from the phase voltage of B and its direction is positive from Y towards B.

If the line voltage between;

- Line 1 and Line 2 = V_{RY}
- Line 2 and Line 3 = V_{YB}
- Line 3 and Line 1 = V_{BR}

Then, we see that V_{RY} leads V_{YB} by 120° and V_{YB} leads V_{BR} by 120° .

Let's suppose,

$$V_{RY} = V_{YB} = V_{BR} = V_L \quad \text{..... (Line Voltage)}$$

Then

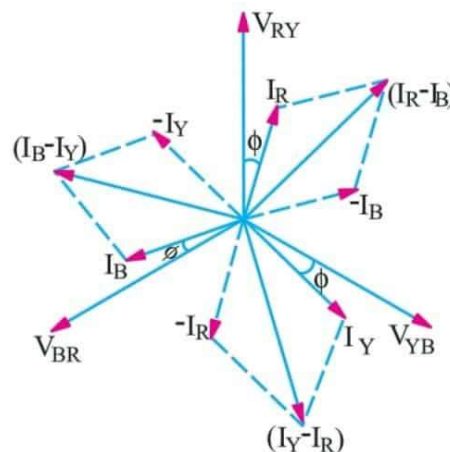
$$V_L = V_{PH}$$

I.e. in Delta connection, the Line Voltage is equal to the Phase Voltage.

Line Currents (I_L) and Phase Currents (I_{Ph}) in Delta Connection

It will be noted from the below (fig-2) that the **total current of each Line is equal to the vector difference between two phase currents in Delta connection** flowing through that line. i.e.;

- Current in Line 1 = $I_1 = I_R - I_B$
- Current in Line 2 = $I_2 = I_Y - I_R$
- Current in Line 3 = $I_3 = I_B - I_Y$ {Vector Difference}



The current of Line 1 can be found by determining the vector difference between I_R and I_B and we can do that by increasing the I_B Vector in reverse, so that, I_R and I_B makes a parallelogram. The diagonal of that parallelogram shows the vector difference of I_R and I_B which is equal to current in Line 1 = I_1 . Moreover, by reversing the vector of I_B , it may indicate as $(-I_B)$, therefore, the angle between I_R and $-I_B$ (I_B , when reversed = $-I_B$) is 60° .

If, $I_R = I_Y = I_B = I_{PH}$ The phase currents

Then;

The current flowing in Line 1 would be;

$$I_L \text{ or } I_1 = 2 \times I_{PH} \times \cos(60^\circ/2)$$

$$= 2 \times I_{PH} \times \cos 30^\circ$$

$$= 2 \times I_{PH} \times (\sqrt{3}/2) \dots\dots \text{Since } \cos 30^\circ = \sqrt{3}/2$$

$$I_L = \sqrt{3} I_{PH}$$

i.e. **In Delta Connection, The Line current is $\sqrt{3}$ times of Phase Current.**

Similarly, we can find the remaining two Line currents as same as above. i.e.,

$$I_2 = I_Y - I_R \dots \text{Vector Difference} = \sqrt{3} I_{PH}$$

$$I_3 = I_B - I_Y \dots \text{Vector difference} = \sqrt{3} I_{PH}$$

As, all the Line current are equal in magnitude i.e.

$$I_1 = I_2 = I_3 = I_L$$

Hence

$$I_L = \sqrt{3} I_{PH}$$

It is seen from the fig above that;

- The Line Currents are 120° apart from each other
- Line currents are lagging by 30° from their corresponding Phase Currents
- The angle Φ between line currents and respective line voltages is $(30^\circ + \Phi)$, i.e. each Line current is lagging by $(30^\circ + \Phi)$ from the corresponding line voltage.

3b. Three impedances each of $(10+j3)$ ohms are connected in star to a 220 V, 3-phase, 50 Hz supply. Calculate the line currents and power delivered to the load. -5M

Ans:

$$3 \text{ } \textcircled{b} \quad Z = (10 + j3) \Omega, \quad V_L = 220V, \quad f = 50 \text{ Hz}.$$

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{220}{\sqrt{3}} = 127.5V$$

$$I_{ph} = \frac{V_{ph}}{Z} = \frac{127.3}{10.44} = 12.19A \approx I_L$$

$$Z = \sqrt{10^2 + 3^2} = 10.44 \Omega$$

$$P = \sqrt{3} V_L I_L \cos \phi$$

$$\phi = \tan^{-1}(3/10) = 16.7^\circ$$

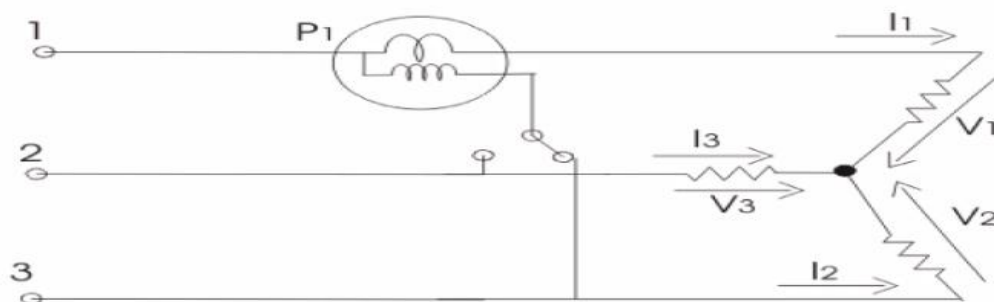
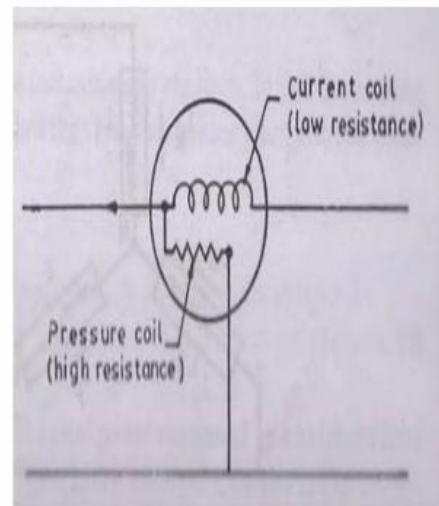
$$P = \sqrt{3} \times 220 \times 12.19 \times \cos(16.7^\circ) = 4703.5W = 4.7035KW$$

4a. Explain the measurement of power in a balanced 3-phase system using a single watt meter. -5M

Ans:

Measurement of power in a balanced 3-phase system using a single watt meter method.

- Power is measured in the electric circuit using a wattmeter.
- A single phase wattmeter consists of two coils; namely the current coil and the pressure coil.
- The Current coil is connected in series with the line and thus carries the line current.
- The Pressure coil is connected in parallel with the line.
- The Wattmeter gives the power per phase.



- The wattmeter gives the value of power per phase.
- Therefore, Total power = 3 X Power per phase
= 3 X wattmeter reading
- The one wattmeter method is used for power measurement in the 3-phase star connected balanced load.

4b. Three impedances each of $(10+j3)$ ohms are connected in star to a 220 V, 3-phase, 50 Hz supply. Calculate the line currents and power delivered to the load. -5M

Ans:

$$Z = (10 + j3) \Omega, \quad V_L = 220V, \quad f = 50 \text{ Hz}$$

$$V_{ph} = \frac{V_L}{\sqrt{3}} = \frac{220}{\sqrt{3}} = 127.5V$$

$$I_{ph} = \frac{V_{ph}}{Z} = \frac{127.5}{10.44} = 12.19A$$

$$P = \sqrt{3} V_L I_L \cos \phi$$

$$\phi = \tan^{-1} \left(\frac{3}{10} \right) = 16.7^\circ$$

$$P = \sqrt{3} \times 220 \times 12.19 \times \cos(16.7^\circ) = 4703.5W = 4.7035KW$$

5. Derive the condition for the maximum power to be transferred from the source to the load with example. - 10M

Ans:

4.8 MAXIMUM POWER TRANSFER

In many practical situations, a circuit is designed to provide power to a load. While for electric utilities, minimizing power losses in the process of transmission and distribution is critical for efficiency and economic reasons, there are other applications in areas such as communications where it is desirable to maximize the power delivered to a load. We now address the problem of delivering the maximum power to a load when given a system with known internal losses. It should be noted that this will result in significant internal losses greater than or equal to the power delivered to the load.

The Thevenin equivalent is useful in finding the maximum power a linear circuit can deliver to a load. We assume that we can adjust the load resistance R_L . If the entire circuit is replaced by its Thevenin equivalent except for the load, as shown in Fig. 4.48, the power delivered to the load is

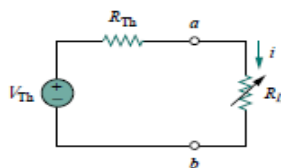


Figure 4.48 The circuit used for maximum power transfer.

$$p = i^2 R_L = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L \quad (4.21)$$

For a given circuit, V_{Th} and R_{Th} are fixed. By varying the load resistance R_L , the power delivered to the load varies as sketched in Fig. 4.49. We notice from Fig. 4.49 that the power is small for small or large values of R_L but maximum for some value of R_L between 0 and ∞ . We now want to show that this maximum power occurs when R_L is equal to R_{Th} . This is known as the *maximum power theorem*.

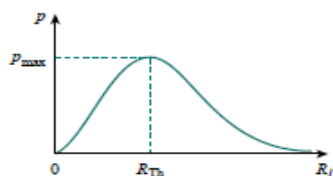


Figure 4.49 Power delivered to the load as a function of R_L .

Maximum power is transferred to the load when the load resistance equals the Thevenin resistance as seen from the load ($R_L = R_{Th}$).

To prove the maximum power transfer theorem, we differentiate p in Eq. (4.21) with respect to R_L and set the result equal to zero. We obtain

$$\begin{aligned} \frac{dp}{dR_L} &= V_{Th}^2 \left[\frac{(R_{Th} + R_L)^2 - 2R_L(R_{Th} + R_L)}{(R_{Th} + R_L)^4} \right] \\ &= V_{Th}^2 \left[\frac{(R_{Th} + R_L - 2R_L)}{(R_{Th} + R_L)^3} \right] = 0 \end{aligned}$$

This implies that

$$0 = (R_{Th} + R_L - 2R_L) = (R_{Th} - R_L) \quad (4.22)$$

which yields

$$R_L = R_{Th} \quad (4.23)$$

showing that the maximum power transfer takes place when the load resistance R_L equals the Thevenin resistance R_{Th} . We can readily confirm that Eq. (4.23) gives the maximum power by showing that $d^2 p / d R_L^2 < 0$.

The maximum power transferred is obtained by substituting Eq. (4.23) into Eq. (4.21), for

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} \quad (4.24)$$

Equation (4.24) applies only when $R_L = R_{Th}$. When $R_L \neq R_{Th}$, we compute the power delivered to the load using Eq. (4.21).

The source and load are said to be matched when $R_L = R_{Th}$.

Any Example: 4M

6a. Explain the procedure to draw the locus diagram of R-L series circuit when L is varying. -5M

Ans:

Series RL circuit with varying Reactance X_L :

Refer to the series RL circuit shown in the figure (a) below with constant R and varying X_L . The current I_L lags behind the applied voltage V by a phase angle $\theta = \tan^{-1}(X_L/R)$ for a given value of R as shown in the figure (b) below. When $X_L = 0$ we can see that the current is maximum equal to V/R and lies along the +ve V axis with phase angle equal to 0° . When X_L is increased from zero to infinity the current gradually reduces from V/R to 0 and phase angle increases from 0° to 90° . As can be seen from the figure, the tip of the current vector traces the path of a semicircle with its diameter along the +ve V axis and on to its right side.

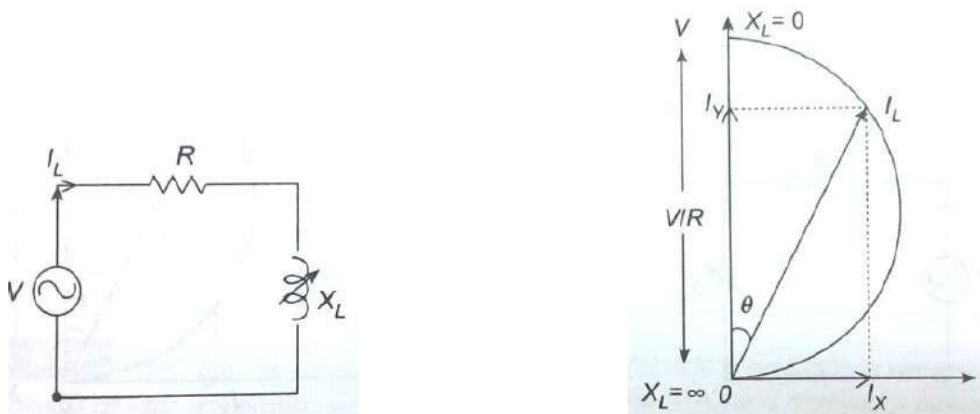


Fig 4.3(a): Series RL circuit with varying X_L Fig 4.3(b) : Locus of current vector I_L with variation of X_L

The Locus of current in a series RL circuit is a semi circle whose radius is $\frac{V}{2R}$ and whose center is $(\frac{V}{2R}, 0)$

6b. Determine the resonant frequency and quality factor of a coil for the series circuit consisting of $R=10\ \Omega$, $L=0.1\text{H}$ and $C=10\ \mu\text{F}$. -5M

Ans:

6 ⑤ Given data
 $R=10\ \Omega$, $L=0.1\text{H}$, $C=10\ \mu\text{F}$

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

$$f_0 = \frac{1}{2\pi\sqrt{(0.1)(10 \times 10^{-6})}} = \underline{\underline{159.15\text{ Hz}}}$$

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$Q = \frac{1}{10} \sqrt{\frac{0.1}{10 \times 10^{-6}}}$$

$$= 10$$

7. A series RLC circuit has to be designed so that it has a bandwidth of 320 Hz and inductance of the coil is 0.2H. It is has to resonate at 350Hz, determine the resistance of coil and capacitance of condenser. If the applied voltage is 150V, determine the voltage across capacitor and coil. -10M

Ans:

⑦ Given
 Bandwidth, $\Delta f = 320\text{ Hz}$
 $f_r \text{ or } f_0 = 350\text{ Hz}$
 $V = 150\text{ V}$
 $L = 0.2\text{ H}$
 $R = ?$ $C = ?$ $V_C = ?$, $V_L = ?$

$$\therefore \text{B.W, } \Delta f = \frac{R}{2\pi L}$$

$$320 = \frac{R}{2\pi \times 0.2}$$

$$\Rightarrow R = \underline{\underline{402.12\ \Omega}}$$

$$f_0 = \frac{1}{2\pi\sqrt{LC}} \Rightarrow C = \frac{1}{(2\pi f_0)^2 L} = \frac{1}{(2\pi \times 350)^2 \times 0.2}$$

$$C = \underline{\underline{1.034\ \mu\text{F}}}$$

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$Z = \sqrt{(402.12)^2 + (439.82 - 453.39)^2}$$

$$Z = 402.35 \Omega$$

$$I = \frac{V}{Z} = \frac{150}{402.35} = 0.373 \text{ A}$$

$$X_L = 2\pi f L = 2\pi \times 50 \times 0.2 = 439.82 \Omega$$

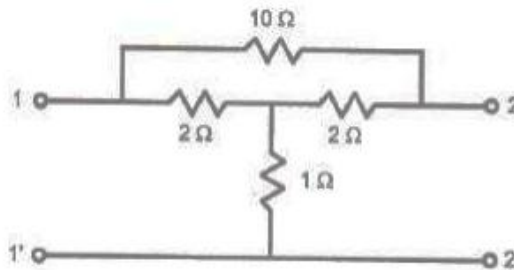
$$X_C = \frac{1}{2\pi f C} = \frac{1}{2\pi \times 50 \times 1.034 \times 10^{-6}} = 453.39 \Omega$$

$$V_L = I X_L = (0.373) \times (439.82) = 164.88 \text{ V}$$

$$V_C = I X_C = (0.373) \times (453.39) = 169.73 \text{ V}$$

8a. Obtain the ABCD parameters of the circuit shown below

-5M



Ans:

8a

Loop 1

$$V_1 = 2(i_1 - i_3) + 1(i_1 + i_2)$$

$$V_1 = 3i_1 + i_2 - 2i_3 \quad \text{--- (1)}$$

Loop 2

$$V_2 = 2(i_2 + i_3) + 1(i_2 + i_1)$$

$$V_2 = i_1 + 3i_2 + 2i_3 \quad \text{--- (2)}$$

Loop 3

$$-10i_3 - 2(i_2 + i_3) - 2(i_3 - i_1) = 0$$

$$2i_1 - 2i_2 - 14i_3 = 0 \quad \text{--- (3)}$$

$$\text{from eq ③ } i_3 = \frac{1}{14} (2i_1 - 2i_2) \\ = \frac{1}{7} (i_1 - i_2) \quad \text{--- ④}$$

Sub eq ④ in eq ①

$$V_1 = 3i_1 + i_2 - 2 \left(\frac{1}{7} (i_1 - i_2) \right) \\ = 3i_1 + i_2 - \frac{2}{7} i_1 + \frac{2}{7} i_2 \\ V_1 = \frac{19}{7} i_1 + \frac{9}{7} i_2 \quad \text{--- ⑤}$$

Sub eq ④ in eq ②

$$V_2 = i_1 + 3i_2 + \frac{2}{7} (i_1 - i_2) = i_1 + 3i_2 + \frac{2}{7} i_1 - \frac{2}{7} i_2 \\ = \frac{9}{7} i_1 + \frac{19}{7} i_2 \quad \text{--- ⑥}$$

$$\frac{9}{7} i_1 = V_2 - \frac{19}{7} i_2$$

$$i_1 = \frac{7}{9} V_2 - \frac{19}{9} i_2 \quad \text{--- ⑦}$$

Compare with

$$I_1 = A V_2 - B I_2 \Rightarrow \boxed{A = \frac{7}{9} \Omega \text{ \& } B = \frac{19}{9}}$$

Sub eq ⑦ in eq ⑤

$$V_1 = \frac{19}{7} i_1 + \frac{9}{7} i_2$$

$$V_1 = \frac{19}{7} \left[\frac{7}{9} V_2 - \frac{19}{9} i_2 \right] + \frac{9}{7} i_2$$

$$V_1 = \frac{19}{9} V_2 - \left[\frac{19}{7} \times \frac{19}{9} - \frac{9}{7} \right] i_2$$

$$= \frac{19}{9} V_2 - \frac{40}{9} i_2 \quad \text{--- ⑧}$$

Compare with

$$V_1 = A V_2 - B I_2$$

$$\boxed{A = \frac{19}{9} \text{ \& } B = \frac{40}{9} \Omega}$$

8b. Derive the relationship between hybrid and Z parameters of two port network. -5M

Ans:

Z-parameters in Terms of Hybrid Parameters We know that

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

Rewriting the second equation,

$$V_2 = -\frac{h_{21}}{h_{22}} I_1 + \frac{1}{h_{22}} I_2$$

Comparing with

$$V_2 = Z_{21} I_1 + Z_{22} I_2,$$

$$Z_{21} = -\frac{h_{21}}{h_{22}}$$

$$Z_{22} = \frac{1}{h_{22}}$$

Also, $V_1 = h_{11} I_1 + h_{12} \left[-\frac{h_{21}}{h_{22}} I_1 + \frac{1}{h_{22}} I_2 \right] = h_{11} I_1 + \frac{h_{12}}{h_{22}} I_2 - \frac{h_{12} h_{21}}{h_{22}} I_1 = \left[\frac{h_{11} h_{22} - h_{12} h_{21}}{h_{22}} \right] I_1 + \frac{h_{12}}{h_{22}} I_2$

Comparing with

$$V_1 = Z_{11} I_1 + Z_{12} I_2,$$

$$Z_{11} = \frac{h_{11} h_{22} - h_{12} h_{21}}{h_{22}} = \frac{\Delta h}{h_{22}}$$

$$Z_{12} = \frac{h_{12}}{h_{22}}$$

Hybrid Parameters in terms of Z-parameters We know that

$$V_1 = Z_{11} I_1 + Z_{12} I_2$$

$$V_2 = Z_{21} I_1 + Z_{22} I_2$$

Rewriting the second equation,

$$I_2 = -\frac{Z_{21}}{Z_{22}} I_1 + \frac{1}{Z_{22}} V_2$$

Comparing with

$$I_2 = h_{21} I_1 + h_{22} V_2,$$

$$h_{21} = -\frac{Z_{21}}{Z_{22}}$$

$$h_{22} = \frac{1}{Z_{22}}$$

Also,

$$V_1 = Z_{11} I_1 + Z_{12} \left[-\frac{Z_{21}}{Z_{22}} I_1 + \frac{1}{Z_{22}} V_2 \right] = \left[\frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{22}} \right] I_1 + \frac{Z_{12}}{Z_{22}} V_2$$

Comparing with

$$V_1 = h_{11} I_1 + h_{12} V_2,$$

$$h_{11} = \frac{Z_{11} Z_{22} - Z_{12} Z_{21}}{Z_{22}} = \frac{\Delta Z}{Z_{22}}$$

$$h_{12} = \frac{Z_{12}}{Z_{22}}$$

9a. Discuss the various types of network topology matrices which are useful for solving any electric circuit or network problem by using their equivalent graphs. -7M

Ans:

INCIDENCE MATRIX

A linear graph is made up of nodes and branches. When a graph is given, it is possible to tell which branches are incident at which nodes and what are its orientations relative to the nodes.

Complete Incidence Matrix (A_a)

For a graph with n nodes and b branches, the complete incidence matrix is a rectangular matrix of order $n \times b$. Elements of this matrix have the following values:

- $a_{ij} = 1$, if branch j is incident at node i and is oriented away from node i .
- $= -1$, if branch j is incident at node i and is oriented towards node i .
- $= 0$, if branch j is not incident at node i .

For the graph shown in Fig. 5.12, branch 1 is incident at nodes 1 and 4. It is oriented away from Node 1 and oriented towards Node 4. Hence, $a_{11} = 1$ and $a_{41} = -1$. Since branch 1 is not incident at nodes 2 and 3, $a_{21} = 0$ and $a_{31} = 0$. Similarly, other elements of the complete incidence matrix are written.

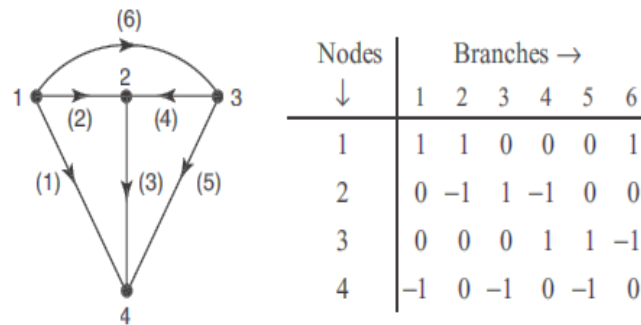


Fig. Graph

The complete incidence matrix is

$$A_a = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & -1 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & -1 \\ -1 & 0 & -1 & 0 & -1 & 0 \end{bmatrix}$$

It is seen from the matrix A_a that the sum of the elements in any column is zero. Hence, any one row of the complete incidence matrix can be obtained by the algebraic manipulation of other rows.

Tie-Set Matrix or Tie- Set Schedule:

When a graph is given, first select a tree and remove all the links. When a link is replaced, a closed loop or circuit is formed. Circuits formed in this way are called fundamental circuits or f -circuits or tiesets.

Orientation of an f -circuit is given by the orientation of the connecting link. The number of f -circuits is same as the number of links for a graph. In a graph having b branches and n nodes, the number of f -circuits or tiesets will be $(b - n + 1)$. Figure 5.15 shows a tree and f -circuits (tiesets) for the graph shown in Fig. 5.14.

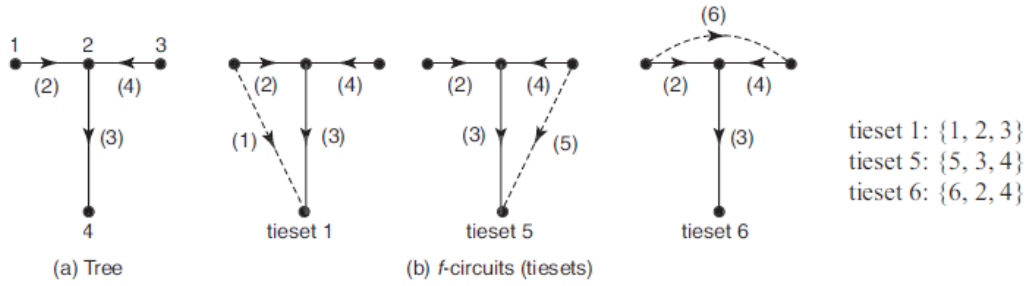


Fig. 5.15 Tree and f -circuits

Here, $b = 6$ and $n = 4$.

$$\text{Number of tiesets} = b - n + 1 = 6 - 4 + 1 = 3$$

f -circuits are shown in Fig. 5.15. The orientation of each f -circuit is given by the orientation of the corresponding connecting link.

The branches 1, 2 and 3 are in the tieset 1. Orientation of tieset 1 is given by orientation of branch 1. Since the orientation of branch 1 coincides with orientation of tieset 1, $b_{11} = 1$. The orientations of branches 2 and 3 do not coincide with the orientation of tieset 1, Hence, $b_{12} = -1$ and $b_{13} = -1$. The branches 4, 5 and 6 are not in tieset 1. Hence, $b_{14} = 0$, $b_{15} = 0$ and $b_{16} = 0$. Similarly, other elements of the tieset matrix are written. Then, the tieset schedule will be written as

Tiesets ↓	Branches →					
	1	2	3	4	5	6
1	1	-1	-1	0	0	0
5	0	0	-1	-1	1	0
6	0	-1	0	1	0	1

Fundamental Cutset and Fundamental Cutset Matrix

When a graph is given, first select a tree and note down its twigs. When a twig is removed from the tree, it separates a tree into two parts (one of the separated part may be an isolated node). Now, all the branches connecting one part of the disconnected tree to the other along with the twig removed, constitutes a cutset. This set of branches is called a fundamental cutset or f -cutset. A matrix formed by these f -cutsets is called an f -cutset matrix. The orientation of the f -cutset is made to coincide with the orientation of the defining branch, i.e., twig. The number of f -cutsets is the same as the number of twigs for a graph.

Figure 5.18 shows a graph, selected tree and f -cutsets corresponding to the selected tree.

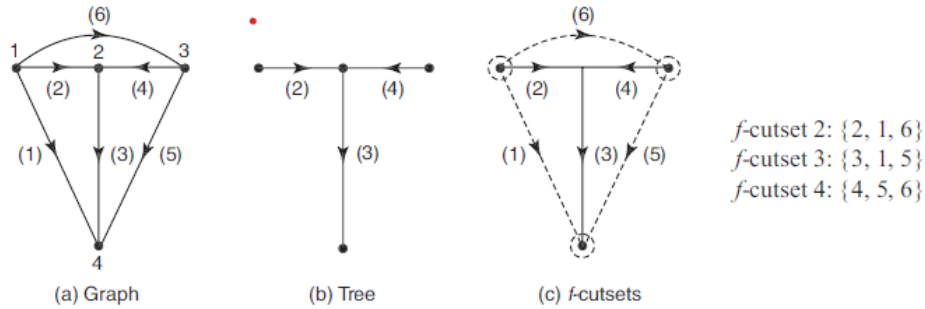


Fig. 5.18 Graph, selected tree and f -cutsets

The branches 2, 1, and 6 are in the f -cutset 2. Orientation of f -cutset 2 is given by orientation of the branch 2 which is moving away from the f -cutset 2. Since the orientations of branches 2, 1 and 6 coincide with the orientation of the f -cutset 2, $q_{11} = 1$, $q_{12} = 1$ and $q_{16} = 0$. The branches 3, 4 and 5 are not in the f -cutset 2. Hence, $q_{13} = 0$, $q_{14} = 0$ and $q_{15} = 0$.

Similarly, other elements of the f -cutset matrix are written.

The cutset schedule is

f -cutsets ↓	Branches →					
	1	2	3	4	5	6
2	1	1	0	0	0	1
3	1	0	1	0	1	0
4	0	0	0	1	1	-1

9b. Outline the principle of duality.

-3M

Ans:

The principle of duality is a fundamental concept in circuit theory and network analysis. It states that there is a one-to-one correspondence between elements of a given electrical network and their duals in a dual network. Essentially, for every element in the original network, there exists a dual element in the dual network, and their behavior is similar in a complementary manner.

Key Features of the Principle of Duality:

1. Duality between Circuit Elements:

- Voltage Source (V) $\leftrightarrow$ Current Source (I).

The duals of each element behave in a complementary manner under duality.

2. Duality in Circuit Laws:

- Kirchhoff's Voltage Law (KVL) $\leftrightarrow$ Kirchhoff's Current Law (KCL):

3. Dual Network Elements:

- Voltage Source $\leftrightarrow$ Current Source:
- Series Combination $\leftrightarrow$ Parallel Combination

Original Network Element	Dual Network Element
Voltage Source (V)	Current Source (I)
Current Source (I)	Voltage Source (V)
Resistor (R)	Inductor (L) or Capacitor (C) (depending on context)
Inductor (L)	Capacitor (C)
Series Combination	Parallel Combination
Parallel Combination	Series Combination
KVL (Sum of Voltages in Loop)	KCL (Sum of Currents at Node)

10a. Explain about the transient response of series RL circuit to the AC excitation for zero initial conditions. **-5M**

Ans:

Derivation of Transient Response in Series R-L Circuit with Sinusoidal Excitation

Let $v = V_m \sin(\omega t + \phi)$ where ϕ varies from 0-2 π depending on the switching instant. Application of KVL at $t = 0^+$ in figure 1 after switch is closed, gives

$$Ri + L \frac{di}{dt} = V_m \sin(\omega t + \phi)$$

$$\text{or, } \frac{R}{L}i + \frac{di}{dt} = \frac{V_m}{L} \sin(\omega t + \phi)$$

...(1)

where “p” represents first derivative of i.

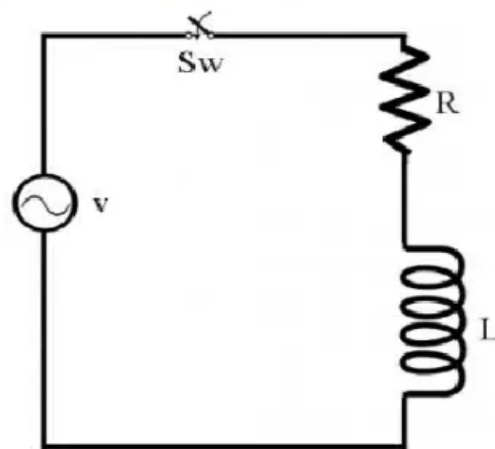


Figure: 1 Series R-L circuit

The complementary function of equation 1 is,

$$i_C = C e^{-(R/L)t} \quad \dots(2)$$

Next we are to obtain the particular solution of current (i_p) such that the net current solution i is given by

$$i = i_C + i_p \text{ forced response}$$

$$\text{Let } i_p = A \cos(\omega t + \phi) + B \sin(\omega t + \phi) \quad \dots(3)$$

where A and B are unknown constants.

∴ The first derivative of i_p becomes

$$p i_p = -A\omega \sin(\omega t + \phi) + B\omega \cos(\omega t + \phi) \quad \dots(4)$$

Substituting these expressions for i_p and $p i_p$ in equation 1 or,

Equating coefficients of like terms,

$$[-A\omega + \frac{BR}{L}] = \frac{V_m}{L}; A = \frac{-\omega LV_m}{R^2 + \omega^2 L^2}$$

$$[B\omega + \frac{AR}{L}] = 0; B = \frac{RV_m}{R^2 + \omega^2 L^2}$$

Substituting the values of A and B in the expression for i_p (equation 3)

or,

$$= \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} [-\omega L \cos(\omega t + \phi) + R \sin(\omega t + \phi)]$$

Let $R = \cos \theta$ and $\omega L = \sin \theta$; θ being $\tan^{-1} (\omega L/R)$,

Thus,

$$\begin{aligned} \text{or, } i_p &= \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \sin(\omega t + \phi - \theta) \\ &= \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} [\sin(\omega t + \phi - \tan^{-1} \frac{\omega L}{R})] \end{aligned}$$

This gives the net current solution as

$$i = i_C + i_p \text{ or, } \dots(5)$$

However, the inductance will oppose any sudden change of current through it.

As there was no current in the circuit before the switch was closed, hence $i_0 = 0$.

Then at $t = 0$.

$$i_0 = 0 = C + \frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \sin(\phi - \tan^{-1} \frac{\omega L}{R})$$

$$\text{or, } C = -\frac{V_m}{\sqrt{R^2 + \omega^2 L^2}} \sin(\phi - \tan^{-1} \frac{\omega L}{R})$$

Substituting the value of C, in equation 5, $\dots(6)$

The final part of equation 6 contains the exponential factor $e^{-(R/L)t}$ which becomes zero at relatively short time as with increasing t, $e^{-(R/L)t}$ decreases. The expression within the bracket depends on the value of ϕ i.e., the instant when the switch is closed.

The second part of equation 5 is the steady state current which lags the applied voltage by an angle given by $\tan^{-1}(\omega L/R)$.

10b. Derive the expression for the current in a series RL circuit ($R = 10\Omega$, $L = 10\text{mH}$) excited by a sinusoidal voltage of 100V, 50 Hz if the supply is connected at $t = 0$. Assume zero initial conditions. -5M

Ans:

10 ⑥ $R = 10\Omega$, $L = 10\text{mH}$, $f = 50\text{Hz}$, $V = 100\text{V}$, $\omega L = 2\pi fL = 100\pi$
 $V(t) = 100 \sin(100\pi t)$

For a RL series circuit,

$$V(t) = R i(t) + L \frac{di(t)}{dt}$$

$$100 \sin(100\pi t) = 10 i(t) + 0.01 \frac{di(t)}{dt}$$

$$10000 \sin(100\pi t) = 1000 i(t) + \frac{di(t)}{dt}$$

$$\frac{di}{dt} + 1000 i(t) = 10000 \sin(100\pi t)$$

Solving the diff. eqn.

The homogeneous eqn is

$$\frac{di_h(t)}{dt} + 1000 i_h(t) = 0$$

$$i_h(t) = C_1 e^{-1000t}$$

The particular solution.

$$i_p(t) = A \cos(100\pi t) + B \sin(100\pi t)$$

Differentiating $i_p(t)$

$$\frac{di_p(t)}{dt} = -A(100\pi) \sin(100\pi t) + B(100\pi) \cos(100\pi t)$$

Sub. $i_p(t)$ and $\frac{di_p(t)}{dt}$ into the original eqn.

$$-100\pi A \sin(100\pi t) + 100\pi B \cos(100\pi t) + 1000[A \cos(100\pi t) + B \sin(100\pi t)] = 10000 \sin(100\pi t)$$

Separating coefficients of $\sin(100\pi t)$ and $\cos(100\pi t)$, we get

$$\sin(100\pi t) \Rightarrow -100\pi A + 1000 B = 10000$$

$$\cos(100\pi t) \Rightarrow 100\pi B + 1000 A = 0$$

Solving A & B

$$A = -3.18, \quad B = 31.8$$

Thus, the particular solution is

$$i_p(t) = -3.18 \cos(100\pi t) + 31.8 \sin(100\pi t)$$

$\therefore$ Total solution is

$$i(t) = i_h(t) + i_p(t)$$

$$= C_1 e^{-1000t} - 3.18 \cos(100\pi t) + 31.8 \sin(100\pi t)$$

Apply initial conditions.

At $t=0$, assume zero initial current, $i(0)=0$

$$i(0) = C_1 e^0 - 3.18 \cos(0) + 31.8 \sin(0) = 0$$

$$C_1 = 3.18$$

The current is

$$i(t) = 3.18 e^{-1000t} - 3.18 \cos(100\pi t) + 31.8 \sin(100\pi t)$$

11. Derive the expression to determine the characteristic impedance of constant K High Pass Filter. -10M

Ans:

9.2 Constant - K High Pass Filters

The constant -K High pass filter as T and π -network are as shown in Fig. 9.21 (a) and (b) respectively. This type of filter can be obtained by changing the positions of series and shunt elements of the constant -K low pass filter.

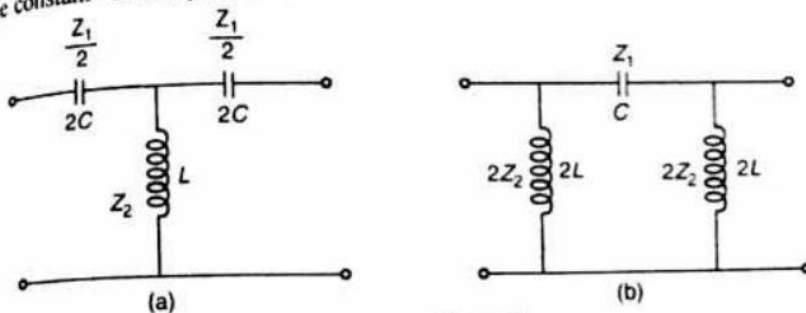


Fig. 9.21

Let $Z_1 = \frac{1}{j\omega C}$ and $Z_2 = j\omega L$, then design impedance for the load impedance is given by

$$R_0^2 = Z_1 Z_2 = \frac{L}{C}$$

$$R_0 = \sqrt{\frac{L}{C}}$$

or

(iv) **Characteristic Impedance:**

The characteristic impedance for T-network is given by

$$Z_{0T} = \sqrt{Z_1 Z_2 \left(1 + \frac{Z_1}{4Z_2}\right)}$$

$$\therefore Z_1 = \frac{1}{j\omega C} \text{ and } Z_2 = j\omega L$$

$$\therefore Z_{0T} = \sqrt{\frac{L}{C} \left(1 - \frac{1}{4\omega^2 LC}\right)}$$

or

$$Z_{0T} = R_0 \sqrt{1 - \left(\frac{f_c}{f}\right)^2}$$

In the pass band, $f > f_c$, so that Z_{0T} is real.

In the stop band, $f < f_c$, so that Z_{0T} is imaginary, and if $f = f_c$, then $Z_{0T} = 0$

The characteristic impedance for π -network is given by

$$Z_{0\pi} = \sqrt{\frac{Z_1 Z_2}{1 + \frac{Z_1}{4Z_2}}}$$

$$= \sqrt{\frac{\frac{L}{C}}{1 - \frac{1}{4\omega^2 LC}}}$$

or

$$Z_{0\pi} = \frac{R_0}{\sqrt{1 - \left(\frac{f_c}{f}\right)^2}}$$

In the pass band, $f > f_c$, so that $Z_{0\pi}$ is real.

In the stop band, $f < f_c$, so that $Z_{0\pi}$ is imaginary, and if $f = f_c$, then $Z_{0\pi} = \infty$

The variation of characteristic impedances Z_{0T} and $Z_{0\pi}$ with frequency f is shown in Fig. 9.24.

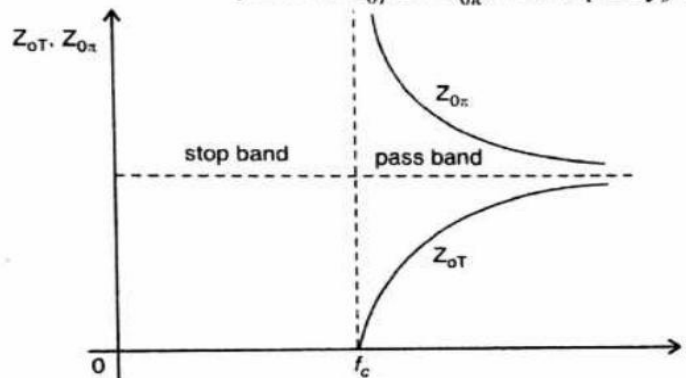


Fig. 9.24